

For loop

control flow statement used to execute a block of code repeatedly based on a condition

It is especially useful when the number of iterations is known in advance, such as iterating over a range of values, arrays, or collections.

Syntax

```
for (initialization; condition; update) {  
    // loop body  
}
```

Example

```
class for loop {  
    public static void main(String[] args) {  
        for (int i = 1; i <= 5; i++) {  
            System.out.println(i);  
        }  
        System.out.println("Loop has ended.");  
    }  
}
```

Explanation

- ❖ The loop variable *i* starts from 1.
- ❖ The loop runs as long as *i* <= 5.
- ❖ After each iteration, *i* is incremented by 1.
- ❖ When the condition becomes false, the loop terminates.